SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL2- COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
Student Name:	D. Gnana Deepthi	**Reg.No.:**	192411123
Department:	CSE	**Date:**	

ANNEXURE A

COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.

2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.

3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.

4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.

5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I D. Gnana Deepthi (192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Gnana Deepthi

ANSWERS

1. Comparison of L1, L2, and L3 Cache Levels

Cache memory is a high-speed memory located close to the CPU. It stores frequently used instructions and data to reduce the time required to access main memory.

Feature	**L1 Cache**	**L2 Cache**	**L3 Cache**
Location	Closest to CPU core	Near the CPU core	Shared among CPU cores
Size	Smallest	Larger than L1	Largest cache level
Latency	Lowest, about 1–5 cycles	Medium, about 10–20 cycles	Higher, about 30–60+ cycles
Bandwidth	Highest	High	Moderate
Throughput	Very high	High	Lower than L1/L2
Cost per bit	Highest	High	Lower than L1/L2
Access frequency	Most frequently used data	Frequently used data not found in L1	Data shared or not found in L1/L2

Impact on Processor Performance

- **L1 Cache:** Because of its extremely low latency and high bandwidth, L1 provides the fastest access to critical data and instructions. A high L1 cache hit rate significantly improves CPU execution speed.
- **L2 Cache:** L2 acts as a backup when data is not available in L1. Although slower than L1, it is much faster than accessing DRAM, helping reduce delays caused by L1 cache misses.
- **L3 Cache:** L3 provides a larger storage area and is often shared between processor cores. It reduces the number of expensive main-memory accesses and improves performance in multicore processors.

2. Difference Between SRAM and DRAM

SRAM (Static Random Access Memory) and DRAM (Dynamic Random Access Memory) are two types of semiconductor memory used at different levels of the computer memory hierarchy.

Feature	SRAM	DRAM
Full Form	Static Random Access Memory	Dynamic Random Access Memory
Storage	Uses flip-flop circuits	Uses capacitor and transistor
Latency	Very low	Higher than SRAM
Bandwidth	Very high	High, but generally lower than SRAM for low-latency access
Throughput	Very high for frequent small accesses	High for large sequential/burst transfers
Refresh	Not required	Required periodically
Density	Low	High
Cost	Expensive	Less expensive
Power	Higher area/power per bit	Lower cost per bit, but refresh consumes power
Typical Use	CPU cache	Main memory (RAM)

SRAM in Cache Design

SRAM is mainly used for **L1, L2, and L3 CPU caches** because it provides very low latency and high-speed access. It does not require refresh operations, allowing the processor to access frequently used instructions and data quickly. Its major disadvantages are high cost and large chip area.

DRAM in Main Memory Design

DRAM is primarily used as **main memory** because it provides much higher storage density at a lower cost per bit. However, its capacitor-based storage requires periodic refreshing, resulting in higher latency compared with SRAM. Modern DRAM can still provide high bandwidth, especially when accessing data in bursts.

3. Comparison of Virtual Memory with Paging and Cache Memory

Virtual memory and cache memory are both memory-management mechanisms, but they work at different levels. **Paging** is a technique used to implement virtual memory, while **cache memory** reduces the time needed to access frequently used data.

Feature	Virtual Memory / Paging	Cache Memory
Purpose	Extends available memory using secondary storage	Speeds up CPU access to frequently used data
Managed by	OS and Memory Management Unit (MMU)	Hardware/cache controller
Storage Used	DRAM + SSD/HDD	SRAM
Access Latency	High, especially during page faults	Very low
Data Unit	Pages, typically several KB	Cache lines, typically tens of bytes
Bandwidth	Lower, especially when swapping to storage	Very high
System Throughput	Can decrease significantly during frequent page faults	Generally increases CPU throughput
Miss Penalty	Very high for a page fault	Relatively low compared with paging

Feature	Virtual Memory / Paging	Cache Memory
Main Benefit	Allows programs to use more memory than physical RAM	Reduces CPU memory-access time

Virtual Memory and Paging

Virtual memory gives each process the illusion of having a large, continuous memory space. **Paging** divides virtual memory and physical memory into fixed-size pages and frames. When a required page is not in RAM, a **page fault** occurs and the operating system must load it from secondary storage.

A page fault has much higher latency than a normal RAM access. Frequent page faults can cause **thrashing**, reducing system throughput because the CPU spends more time waiting for memory transfers.

Cache Memory

Cache memory stores frequently accessed instructions and data close to the CPU. When the required data is found in the cache (**cache hit**), it can be accessed with very low latency. A cache miss requires accessing a lower cache level or main memory, but the penalty is generally much smaller than a virtual-memory page fault.

4. Differences Between NAND Flash and DRAM

NAND Flash and DRAM are both important memory technologies, but they serve different purposes in a hierarchical memory system. DRAM is used mainly as **main memory**, while NAND Flash is commonly used for **secondary storage**.

Feature	NAND Flash	DRAM
Type	Non-volatile memory	Volatile memory
Latency	High	Low
Bandwidth	Moderate to high, especially with parallel channels	High
Random Access	Relatively slow	Very fast
Data Retention	Retains data without power	Loses data when power is removed

Feature	NAND Flash	DRAM
Refresh Required	No	Yes
Capacity	Very high	Moderate to high
Cost per bit	Low	Higher than NAND Flash
Typical Use	SSDs, USB drives, memory cards	Main memory
Position in Hierarchy	Secondary/tertiary storage	Main memory

NAND Flash

NAND Flash is **non-volatile**, meaning it retains information even when power is turned off. It provides high storage density at a relatively low cost. However, its access latency is much higher than DRAM, particularly for random accesses. Modern SSDs improve NAND performance using multiple flash channels, controllers, and parallelism.

DRAM

DRAM provides much lower latency and high bandwidth, making it suitable for directly supplying data to the CPU. It is volatile and requires periodic refresh operations. DRAM is more expensive per bit than NAND Flash but provides significantly better performance for active program data.

Suitability for Hierarchical Memory

A suitable memory hierarchy can be represented as:

CPU Cache → DRAM → NAND Flash/SSD

- **Cache:** Stores the most frequently accessed data and provides extremely low latency.
- **DRAM:** Stores currently running programs and active data with low latency and high bandwidth.
- **NAND Flash:** Provides large, persistent storage at lower cost but with higher access latency.

5. Comparison of MRAM and RRAM Technologies

MRAM (Magnetoresistive Random Access Memory) and **RRAM (Resistive Random Access Memory)** are non-volatile memory technologies that can provide fast access while retaining data without power.

Feature	MRAM	RRAM
Full Form	Magnetoresistive RAM	Resistive RAM
Storage Principle	Stores data using magnetic states	Stores data by changing resistance of a material
Access Time	Very low, typically comparable to fast SRAM	Low, but can vary with device technology
Read Speed	Very fast	Fast
Write Speed	Fast and consistent	Fast, but may vary depending on technology
Throughput	High	High
Energy Efficiency	High, especially because it requires no refresh	Very high due to low-energy switching
Volatility	Non-volatile	Non-volatile
Endurance	Very high	Generally high, but can vary
Typical Applications	Embedded memory, cache-like applications, industrial systems	Embedded memory, storage-class memory, neuromorphic systems

MRAM

MRAM stores information using **magnetic states** rather than electrical charge. It provides fast read/write operations, high endurance, and does not require refresh cycles. Therefore, it can offer good performance with relatively low standby power.

MRAM is suitable for applications requiring **fast and reliable non-volatile memory**, such as embedded systems and high-performance memory applications.

RRAM

RRAM stores data by changing the **electrical resistance** of a material. It has a simple cell structure and offers high density and low-energy operation. Its access time can be very low, and its scalability makes it attractive for future memory systems.

RRAM is particularly promising for **high-density storage, embedded memory, and AI/neuromorphic computing** applications.

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear	Shows reasonable interpretatio	Basic interpretation; lacks depth	No meaningful interpretation

		interpretation and reasoning	n with minor gaps		
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand